

Detection and Threat Services

GuardDuty, Inspector, Macie, Detective and Security Hub. Five services, five jobs.

Every service in this module **tells you**
something is wrong. None of them **stops** it.

The five services, one line each

SERVICE	THE ONE THING IT DOES
Amazon GuardDuty	Watches behaviour for active threats
Amazon Inspector	Scans workloads for software vulnerabilities
Amazon Macie	Finds sensitive data inside Amazon S3
Amazon Detective	Investigates a finding after it appears
AWS Security Hub	Aggregates findings and checks against standards

ON THE EXAM

Behaviour, vulnerabilities, data, investigation, aggregation. Five nouns that separate five services.

GuardDuty or Inspector?

AMAZON GUARDDUTY

Something is happening now

- Reads logs and network activity continuously
- Unusual API calls, crypto-mining, compromised credentials
- Finds an attack in progress
- No agent to install

AMAZON INSPECTOR

Something could be exploited

- Scans EC2, container images and Lambda functions
- Known software vulnerabilities and unintended network exposure
- Finds weaknesses before anyone uses them
- Automatically discovers what to scan

THE TIEBREAKER

An attack in progress means GuardDuty. An unpatched weakness nobody has touched yet means Inspector.

Amazon GuardDuty in detail

- ◆ **It reads logs you already have.** CloudTrail, VPC Flow Logs and DNS logs, without you enabling them separately.
- ◆ Nothing to install. No agent, no appliance, no change to your workloads.
- ◆ **Typical findings:** crypto-mining, credential use from an unusual location, reconnaissance, communication with known-bad addresses.
- ◆ Optional protections extend it to S3, EKS, RDS login activity, Lambda and malware scanning.
- ◆ It never inspects the contents of your data. Only behaviour around it.

ON THE EXAM

Unusual API calls is the single most common giveaway phrase for GuardDuty on this exam.

Amazon Inspector in detail

- ◆ **Amazon EC2 instances**, for unpatched operating system and application vulnerabilities.
- ◆ **Container images in Amazon ECR**, scanned when pushed and rescanned as new vulnerabilities appear.
- ◆ **AWS Lambda functions**, including their code and dependencies.
- ◆ It discovers eligible workloads automatically rather than waiting to be pointed at them.
- ◆ It also reports unintended network exposure, meaning paths reachable from the internet.

ON THE EXAM

If the question mentions CVEs, patching, or vulnerable software, it is Inspector. Nothing else in AWS scans for those.

Macie or GuardDuty?

AMAZON MACIE

What the data is

- Machine learning and pattern matching over Amazon S3
- Discovers personally identifiable information
- Classifies and flags sensitive content at rest
- The only service here that reads your data

AMAZON GUARDDUTY

What is being done

- Threat detection across account activity
- Unusual API calls and network behaviour
- Flags compromise, not content
- Never looks inside your objects

THE TIEBREAKER

Sensitive data, PII, or S3 contents means Macie. Macie is the only service in this module that opens your files.

GuardDuty or Detective?

AMAZON GUARDDUTY

Raises the alarm

- Produces findings on its own
- Tells you something suspicious happened
- The starting point of an investigation
- Runs continuously in the background

AMAZON DETECTIVE

Explains the alarm

- Consumes findings rather than producing them
- Builds a linked view of related activity
- Used to determine root cause and scope
- You open it because something already fired

THE TIEBREAKER

Detective never tells you that something is wrong. It tells you why, after something else already did.

AWS Security Hub

- ◆ **It collects findings from the others** into one place: GuardDuty, Inspector, Macie, IAM Access Analyzer, Firewall Manager and partner tools.
- ◆ **It runs automated best-practice checks** against standards such as CIS, PCI DSS and the AWS Foundational Security Best Practices.
- ◆ It works across many accounts and Regions, which is usually the reason a question reaches for it.
- ◆ Aggregation is not automatically the answer. It is right only when the requirement is a single central view.
- ◆ Findings can be routed onwards with Amazon EventBridge to trigger a response.

ON THE EXAM

One central view across accounts means Security Hub. A specific kind of finding means the service that produces it.

There are now two Security Hubs

- ◆ **At re:Inforce in June 2025, AWS renamed the original service** to AWS Security Hub CSPM.
- ◆ **A different, new service took the name AWS Security Hub** and reached general availability on 2 December 2025.
- ◆ **The exam means the old one.** Findings aggregation plus standards checks is Security Hub CSPM today.
- ◆ The new Security Hub adds risk prioritisation, exposure findings and attack path visualisation on top.
- ◆ They are separate services and their finding formats differ, so they are not interchangeable.

ON THE EXAM

Answer AWS Security Hub on the exam. Recognise Security Hub CSPM in the console. Module 22 covers the wider gap.

Reverse lookup

IF THE SCENARIO SAYS	THE SERVICE IS
Unusual API calls, unexpected behaviour, compromised credentials	Amazon GuardDuty
Unpatched software, vulnerabilities, CVEs, container images	Amazon Inspector
Personally identifiable information, sensitive data in S3	Amazon Macie
Investigate the root cause of an existing finding	Amazon Detective
One view of security findings across many accounts	AWS Security Hub
Resources shared externally, or unused permissions	IAM Access Analyzer

ON THE EXAM

Six services, six phrases. The exam reuses this vocabulary almost word for word.

How a finding travels

01

Something detects

GuardDuty, Inspector or Macie produces a finding.

02

Security Hub collects

Findings from all sources land in one place, in one format.

03

EventBridge routes

A finding becomes an event that can trigger something.

04

You respond

Notify a team, run a Lambda function, or open Detective to investigate.

ON THE EXAM

Detection alone changes nothing. The value comes from routing findings to a response.

Detection or protection?

- ◆ **Detection services tell you.** GuardDuty, Inspector, Macie, Detective, Security Hub.
- ◆ **Protection services stop it.** AWS WAF, AWS Shield, security groups, network ACLs.
- ◆ Answering GuardDuty to a question about blocking an attack is a common and costly mistake.
- ◆ Read the verb in the question. Identify, find, discover, alert means detection.
- ◆ Block, filter, prevent, restrict, protect means the next module.

ON THE EXAM

The verb in the question does most of the work. Detection verbs and protection verbs rarely overlap.

A company's monitoring shows that one of its EC2 instances has begun making outbound connections to an address associated with cryptocurrency mining, and an IAM access key is being used from a country where the company has no staff.

The security team wants a service that would have identified both of these automatically, with nothing to install on the instances.

Which service meets this requirement?

- | | |
|---|---|
| ☐ A Amazon Inspector | ☐ B Amazon Macie |
| ☐ C Amazon GuardDuty | ☐ D AWS Config |

◆ PAUSE AND DECIDE

Which service meets this requirement?

A Amazon Inspector

Finds vulnerabilities that could be exploited. It does not watch behaviour, and nothing here is about a missing patch.

B Amazon Macie

Looks inside S3 for sensitive data. No data content is involved in either symptom.

C Amazon GuardDuty

Crypto-mining traffic and credential use from an unusual location are two of its headline finding types. Agentless.

D AWS Config

Records configuration and evaluates compliance rules. It has no view of network behaviour or credential use.

THE SIGNAL

Two behaviours, no configuration change and no vulnerability. Plus nothing to install, which is a GuardDuty hallmark.

A healthcare company stores several million objects in Amazon S3. It must identify which buckets contain patient records with personally identifiable information, and classify them, in order to satisfy a regulator.

It does not currently know which buckets hold regulated data.

Which service should the company use?

- | | |
|---|---|
| ☐ A Amazon GuardDuty | ☐ B Amazon Macie |
| ☐ C AWS Security Hub | ☐ D Amazon Inspector |

◆ PAUSE AND DECIDE

Which service should the company use?

A Amazon GuardDuty

Would tell them if a bucket were being accessed suspiciously. It cannot tell them what is inside one.

C AWS Security Hub

Aggregates findings that other services produce. With Macie switched off there is nothing to aggregate.

B Amazon Macie

Machine learning and pattern matching over S3 to discover and classify sensitive data. Exactly this requirement.

D Amazon Inspector

Scans compute workloads for vulnerabilities. S3 is not a compute workload and this is not a vulnerability.

THE SIGNAL

Personally identifiable information plus S3 is the Macie signature. Only Macie reads object contents.

A company runs a public web application. Its security team receives an alert that the application is being probed for SQL injection from a range of addresses, and wants to stop those requests reaching the application.

Separately, the team wants to be alerted in future when any AWS credential is used from an unexpected location.

Which two services meet these requirements? (Choose two.)

- | | |
|---|---|
| ☐ A Amazon GuardDuty | ☐ B AWS WAF |
| ☐ C Amazon Inspector | ☐ D Amazon Detective |
| ☐ E AWS Security Hub | |

◆ PAUSE AND DECIDE

Which two services meet these requirements? (Choose two.)

- A

 Amazon GuardDuty
Credential use from an unexpected location is a core GuardDuty finding. Meets the second requirement.
- B

 AWS WAF
Inspects HTTP requests and blocks SQL injection patterns before they reach the application.
- C

 Amazon Inspector
Might report that the application has a vulnerability, but it cannot block a request.
- D

 Amazon Detective
Investigates findings after the fact. It neither blocks nor alerts.
- E

 AWS Security Hub
Would show both findings in one place, but produces neither and blocks nothing.

THE SIGNAL

One requirement uses a protection verb, stop, and one uses a detection verb, alerted. Two different halves.

A GuardDuty finding reports that an EC2 instance has been communicating with a known command-and-control address. The security team needs to establish how the instance was reached, which credentials were used, whether other resources were touched, and over what period.

They want a service that assembles this picture from existing log data.

Which service should they use?

- | | |
|---|--|
| ☐ A Enable a second GuardDuty detector | ☐ B Amazon Detective |
| ☐ C Amazon Inspector | ☐ D AWS Trusted Advisor |

◆ PAUSE AND DECIDE

Which service should they use?

A Enable a second GuardDuty detector
GuardDuty already produced the finding. Running it twice adds nothing; the question is about understanding, not detecting.

C Amazon Inspector
Might later explain how the instance was vulnerable, but it cannot reconstruct what happened.

B Amazon Detective
Builds a linked view from CloudTrail, VPC Flow Logs and GuardDuty findings to establish scope and root cause.

D AWS Trusted Advisor
Best-practice checks on the account. It has no role in incident investigation.

THE SIGNAL

The finding already exists. Once the question moves from what happened to why, it is Detective.

An organisation runs forty AWS accounts. Each account has GuardDuty, Inspector and Macie enabled, and the security team currently reviews findings by signing in to each account in turn.

The team wants one view of all findings, and wants to measure the estate against the CIS benchmark.

Which service meets both requirements?

- | | |
|---|--|
| ☐ A Amazon Detective | ☐ B AWS CloudTrail |
| ☐ C AWS Security Hub | ☐ D Amazon CloudWatch |

◆ PAUSE AND DECIDE

Which service meets both requirements?

A Amazon Detective

Investigates individual findings in depth. It is not a cross-account dashboard.

C AWS Security Hub

Aggregates findings across accounts and runs standards checks including CIS. Both requirements, one service.

B AWS CloudTrail

Records API activity. It does not collect security findings or run benchmark checks.

D Amazon CloudWatch

Metrics, logs and alarms for operational health. It is not a security findings aggregator.

THE SIGNAL

Many accounts plus a named compliance standard is the Security Hub signature. Both halves point the same way.

What to carry out of this module

- ◆ **Everything here detects. Nothing here protects.** Read the verb in the question.
- ◆ **Behaviour is GuardDuty. Vulnerabilities are Inspector. Data content is Macie.**
- ◆ Detective explains a finding that already exists. It never raises one.
- ◆ Security Hub is right when the requirement is one view across accounts, or a named standard.
- ◆ Aggregation is wrong when nothing is producing findings yet. Turn on the source first.

ON THE EXAM

In the console the aggregation service is now called Security Hub CSPM. On the exam, answer Security Hub.